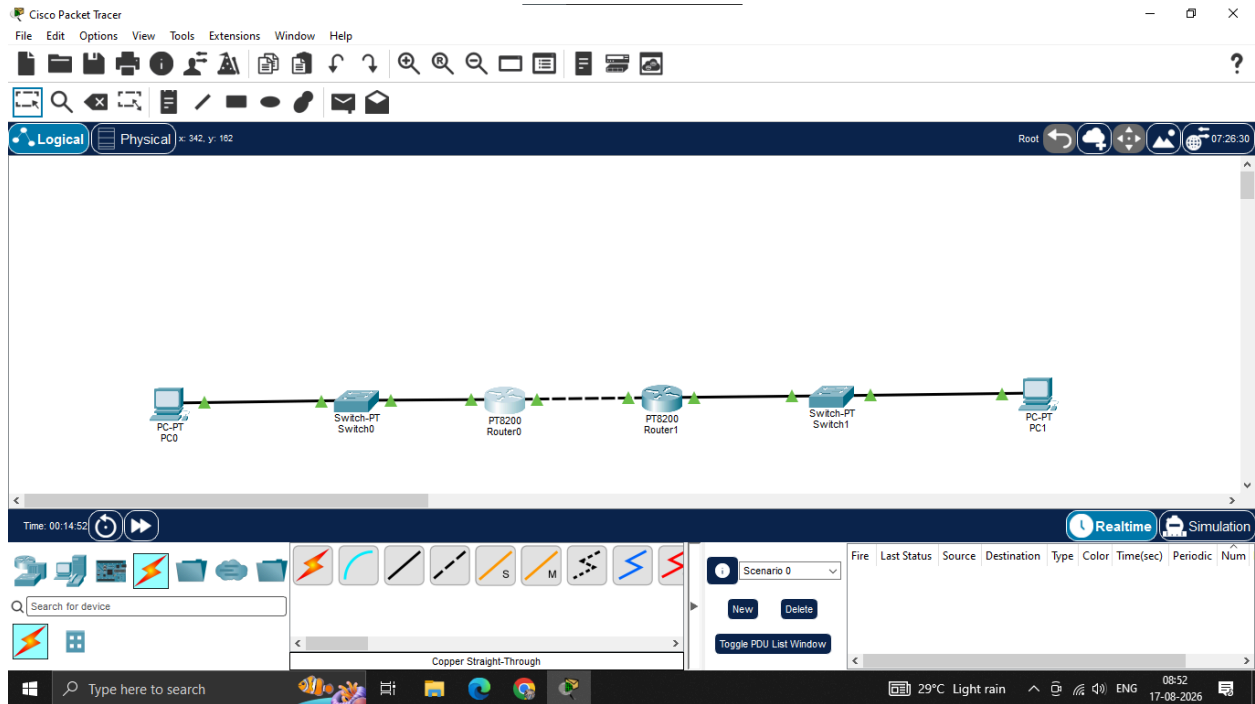


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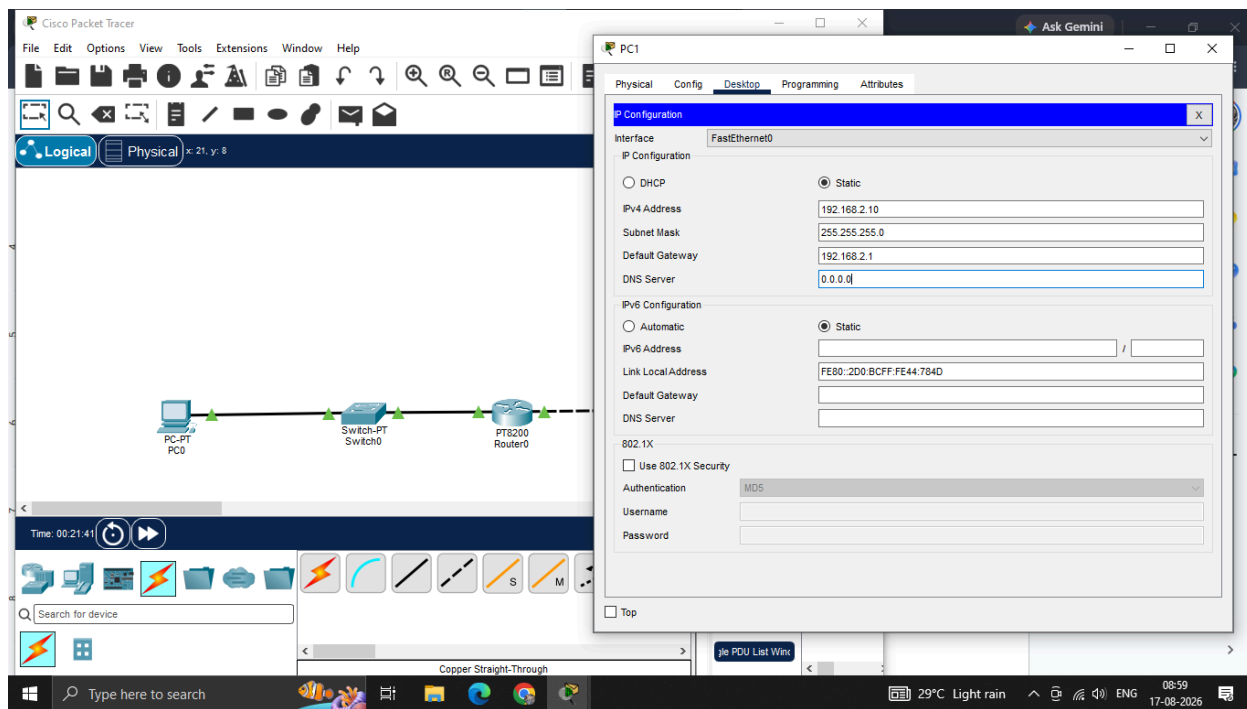
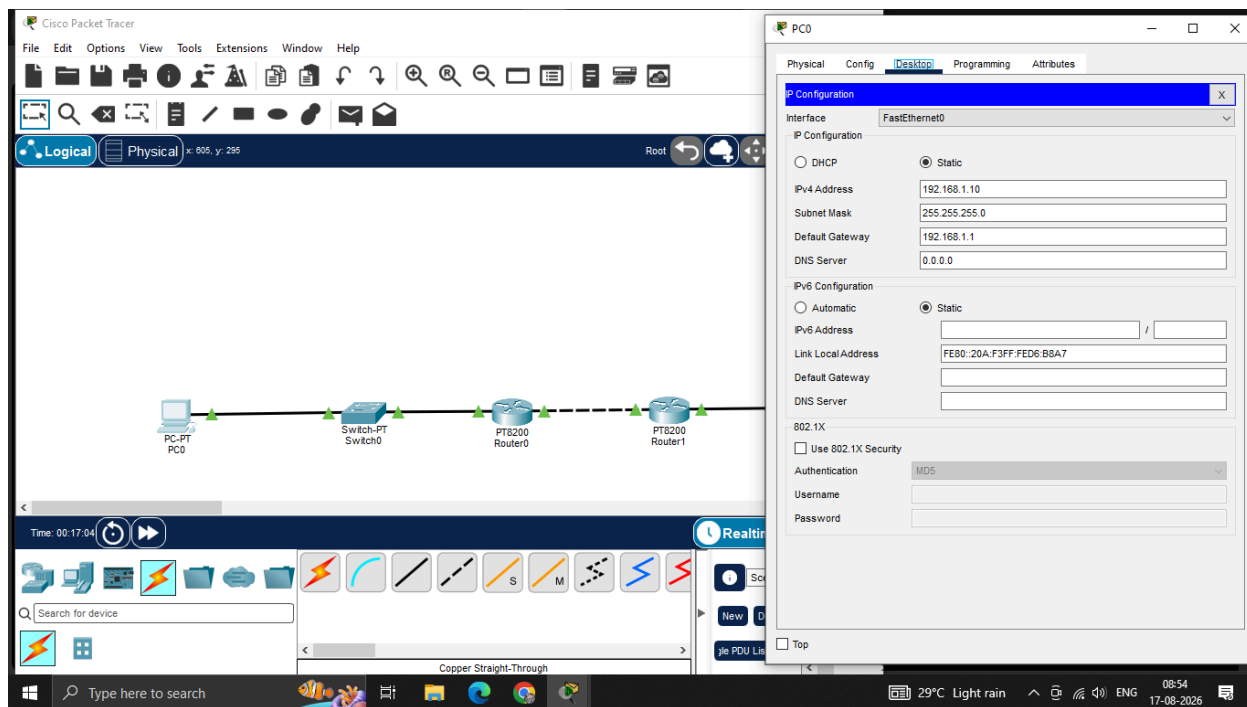
PRACTICAL NO:- 06

AIM :- Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.



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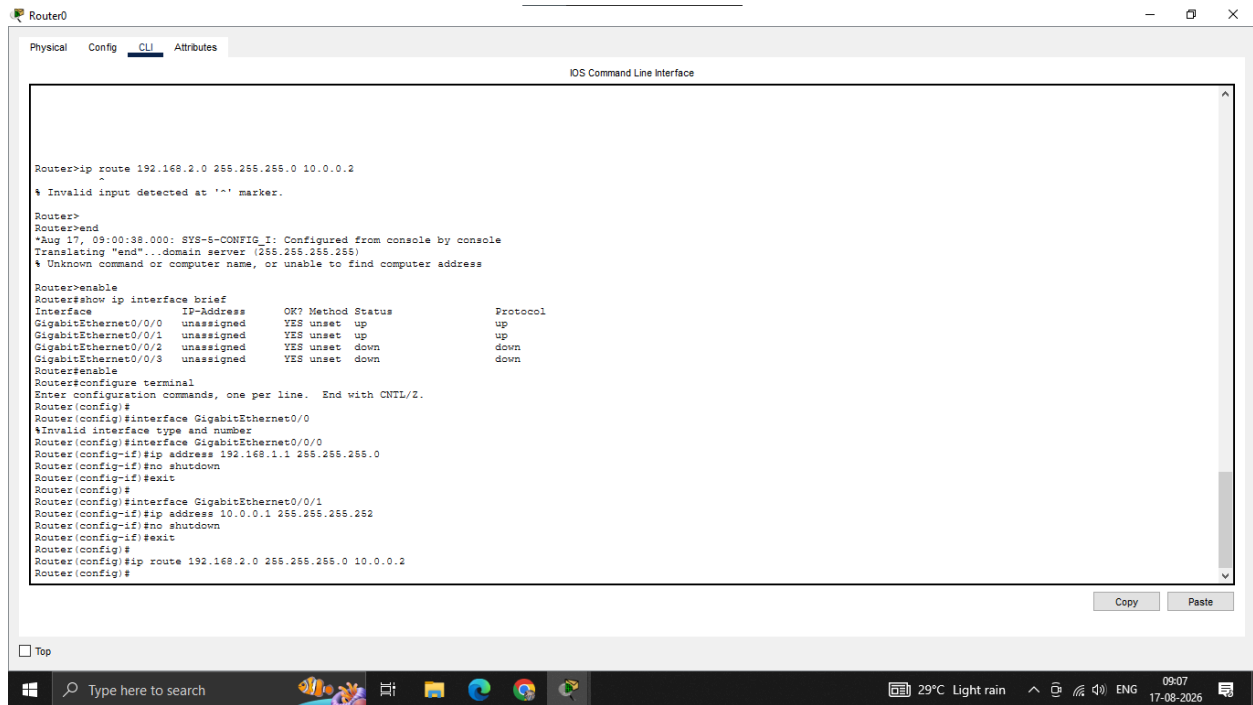
PRACTICAL NO:- 06



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PRACTICAL NO:- 06

Router0 configuration



The screenshot shows the Router0 configuration window with the CLI tab selected. The command history shows the following sequence of commands and their outputs:

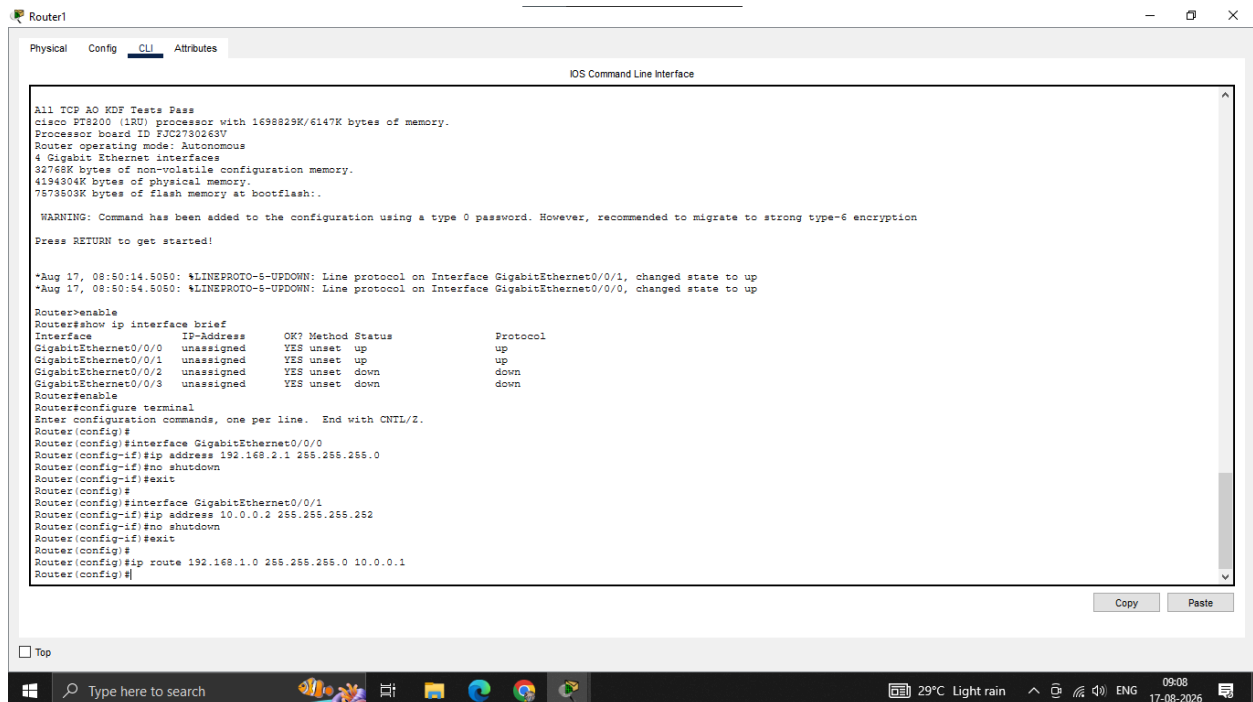
```
Router>ip route 192.168.2.0 255.255.255.0 10.0.0.2
^
% Invalid input detected at '^' marker.

Router>
Router>end
*Aug 17, 09:00:38.000: SYS-5-CONFIG_I: Configured from console by console
Translating "end"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router>enable
Router>show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0  unassigned     YES unset  up          up
GigabitEthernet0/0/1  unassigned     YES unset  up          up
GigabitEthernet0/0/2  unassigned     YES unset  down        down
GigabitEthernet0/0/3  unassigned     YES unset  down        down

Router>enable
Router>configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface GigabitEthernet0/0
%Invalid interface type and number
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#
```

Router1 configure



The screenshot shows the Router1 configuration window with the CLI tab selected. The command history shows the following sequence of commands and their outputs:

```
All TCP AO KDF Tests Pass
cisco PT8200 (1RU) processor with 1698829K/6147K bytes of memory.
Processor board ID FJC2730263V
Router operating mode: Autonomous
4 Gigabit Ethernet interfaces
32768K bytes of non-volatile configuration memory.
4194904K bytes of physical memory.
7873503K bytes of flash memory at bootflash:.

WARNING: Command has been added to the configuration using a type 0 password. However, recommended to migrate to strong type-6 encryption

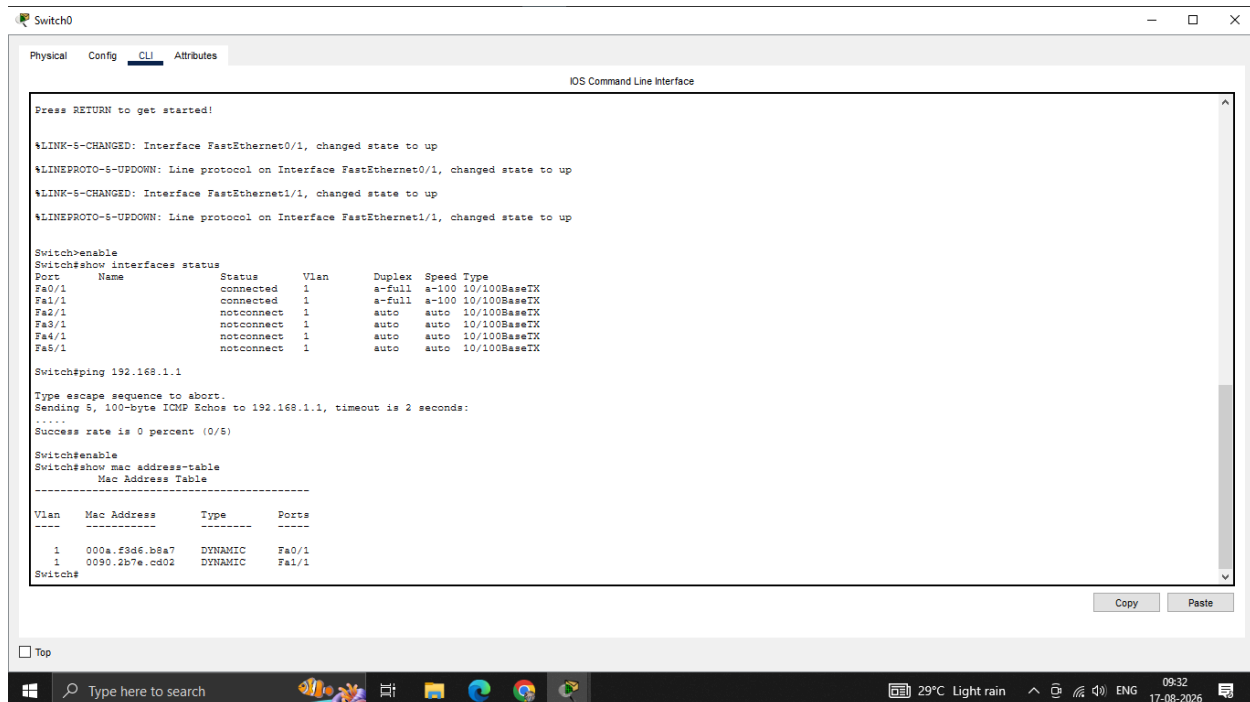
Press RETURN to get started!

*Aug 17, 08:50:14.5050: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
*Aug 17, 08:50:54.5050: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router>enable
Router>show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0  unassigned     YES unset  up          up
GigabitEthernet0/0/1  unassigned     YES unset  up          up
GigabitEthernet0/0/2  unassigned     YES unset  down        down
GigabitEthernet0/0/3  unassigned     YES unset  down        down

Router>enable
Router>configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#
```

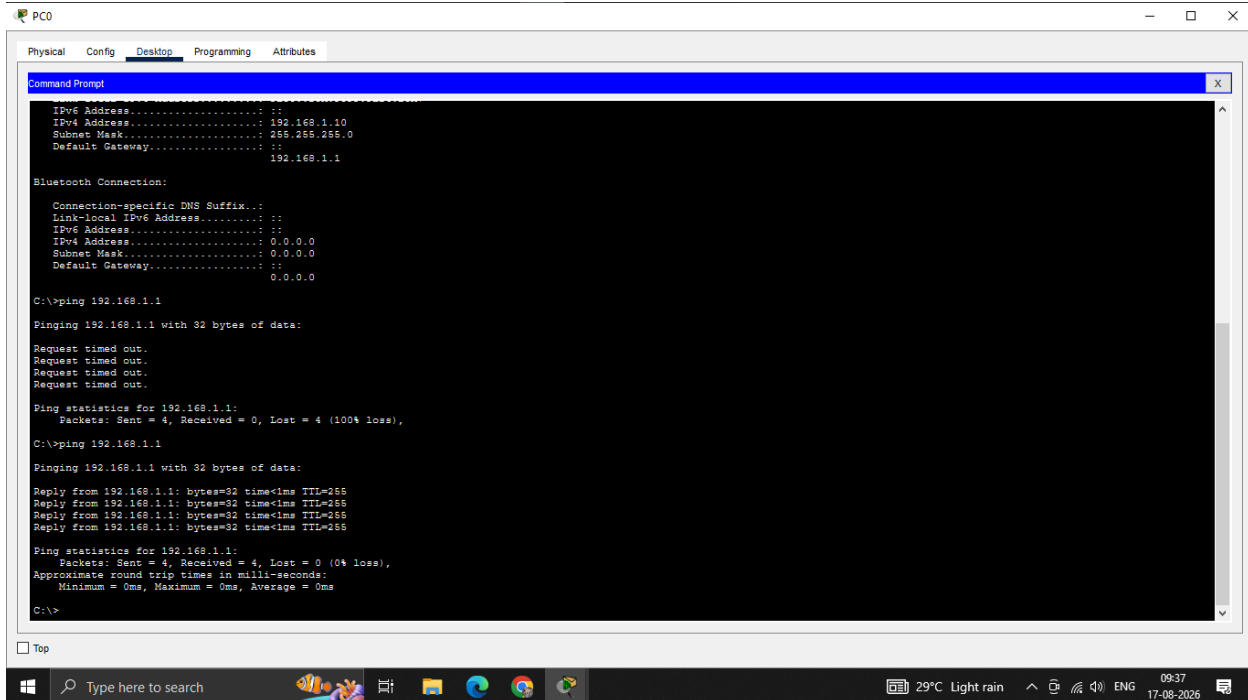
Switch0



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PRACTICAL NO:- 06

PC0 → Router0 connectivity working



The screenshot shows a Windows PC0 window with a Command Prompt open. The Command Prompt displays the following output:

```
IPv6 Address..... ::
IPv4 Address..... 192.168.1.10
Subnet Mask..... 255.255.255.0
Default Gateway..... 192.168.1.1

Bluetooth Connection:
Connection-specific DNS Suffix...:
Link-local IPv6 Address..... ::
IPv6 Address..... ::
IPv4 Address..... 0.0.0.0
Subnet Mask..... 0.0.0.0
Default Gateway..... 0.0.0.0

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

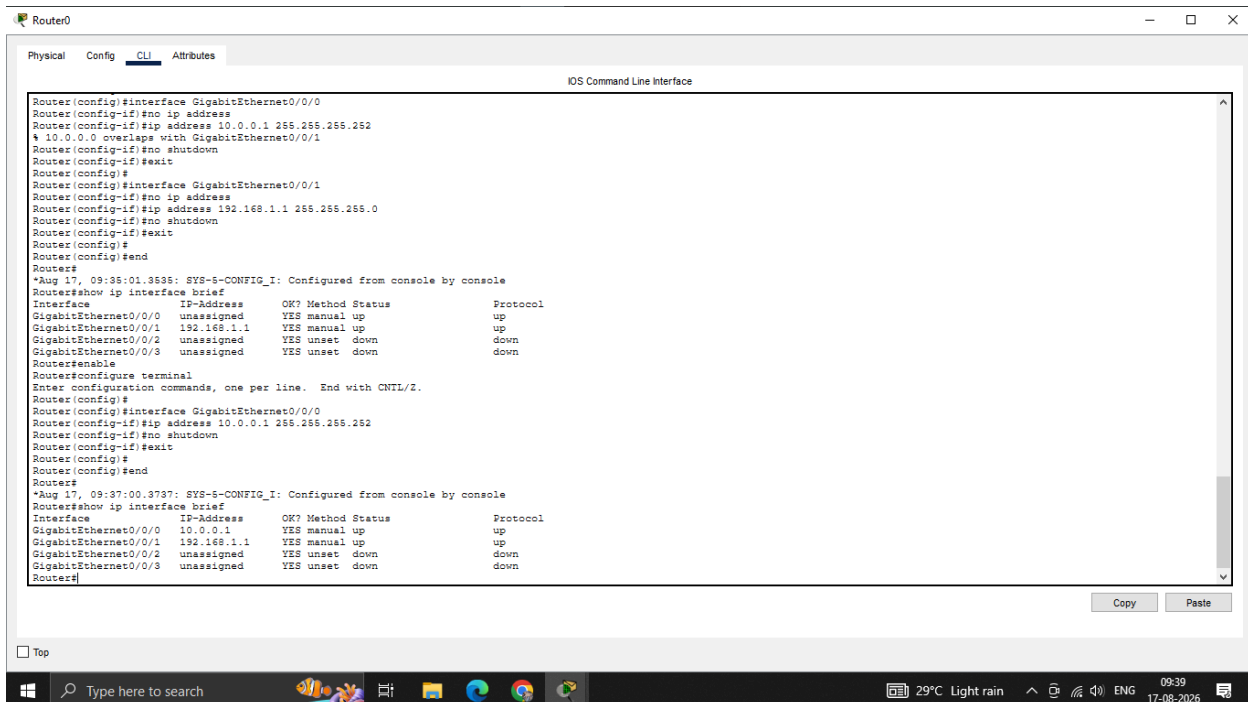
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=4ms TTL=255
Reply from 192.168.1.1: bytes=32 time=4ms TTL=255
Reply from 192.168.1.1: bytes=32 time=4ms TTL=255
Reply from 192.168.1.1: bytes=32 time=4ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```



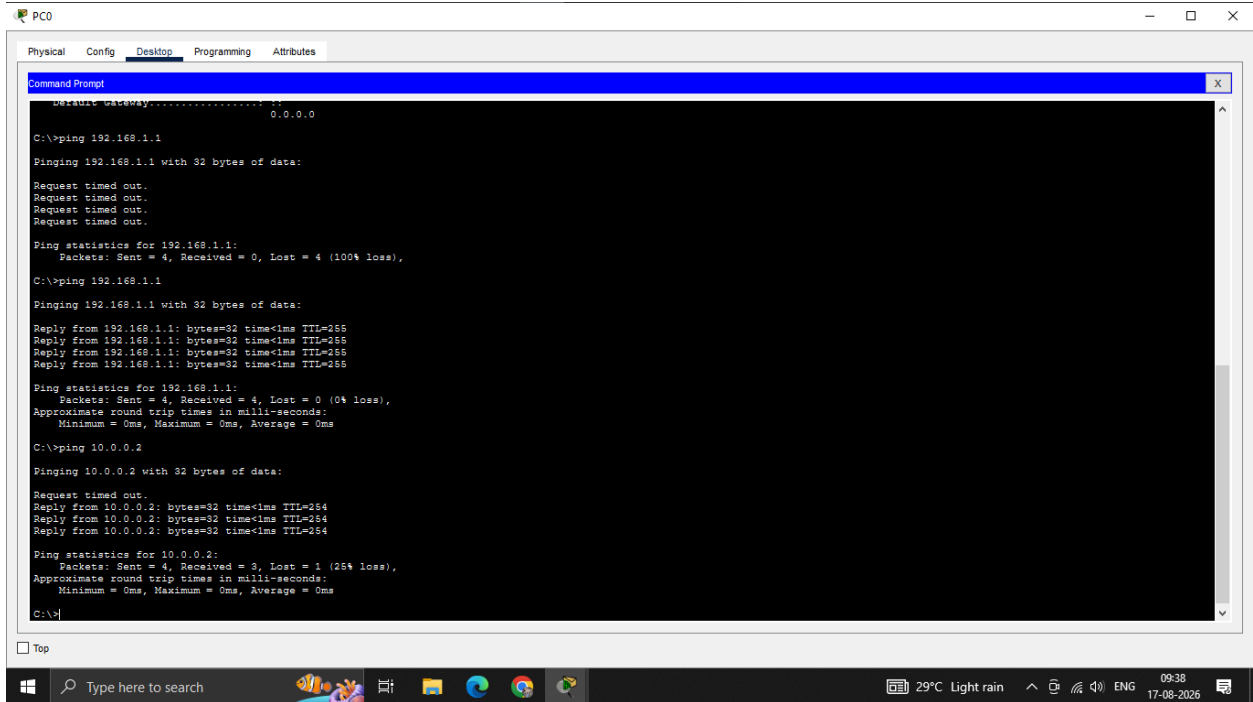
The screenshot shows a Router0 window with the IOS Command Line Interface. The configuration commands and output are as follows:

```
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address
Router(config-if)#ip address 10.0.0.1 255.255.255.252
% 10.0.0.0 overlaps with GigabitEthernet0/0/1
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#no ip address
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#end
Router#
*Aug 17, 09:35:01.3535: SYS-5-CONFIG_I: Configured from console by console
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0  unassigned      YES manual up          up
GigabitEthernet0/0/1  192.168.1.1     YES manual up          up
GigabitEthernet0/0/2  unassigned      YES unset down        down
GigabitEthernet0/0/3  unassigned      YES unset down        down
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#end
Router#
*Aug 17, 09:37:00.3737: SYS-5-CONFIG_I: Configured from console by console
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0  10.0.0.1         YES manual up          up
GigabitEthernet0/0/1  192.168.1.1     YES manual up          up
GigabitEthernet0/0/2  unassigned      YES unset down        down
GigabitEthernet0/0/3  unassigned      YES unset down        down
Router#
```

SHETH L.U.J AND SIR MV COLLEGE

PRACTICAL NO:- 06

Router0 ↔ Router1 connectivity working



The screenshot shows a PC0 desktop environment. The 'Desktop' tab is selected in the top menu. A 'Command Prompt' window is open, displaying the following text:

```
Default gateway: 0.0.0.0

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.2

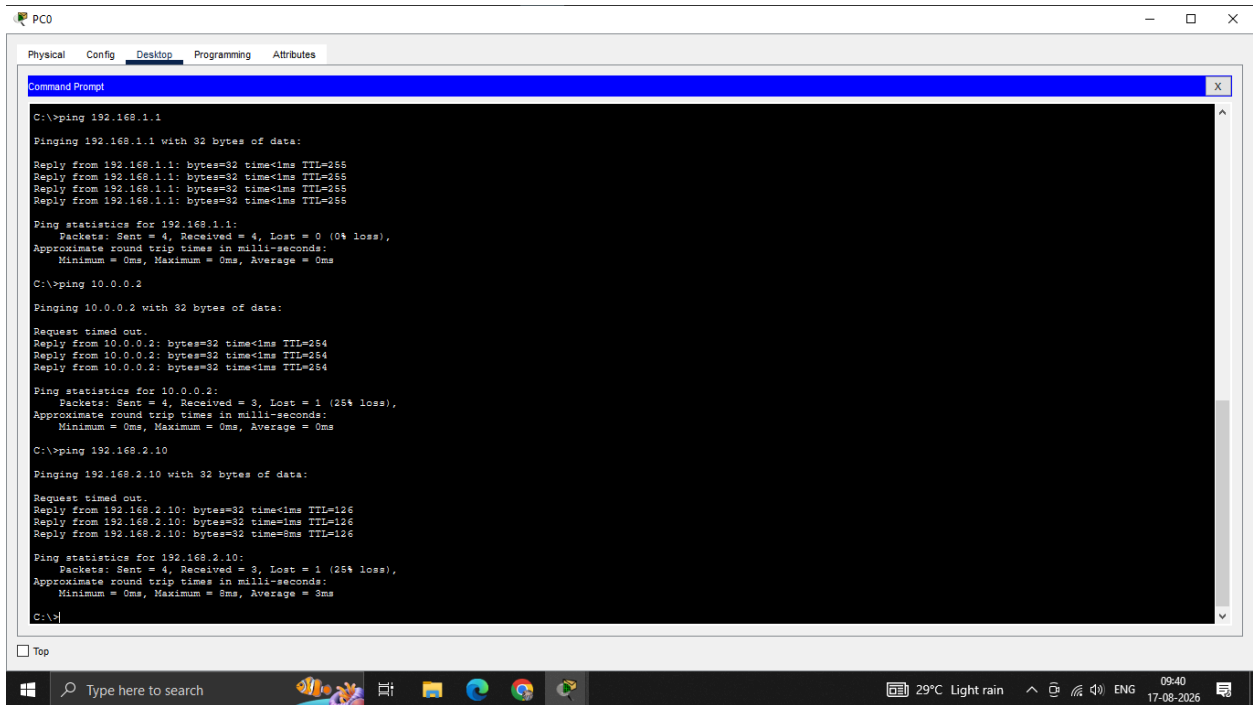
Pinging 10.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 10.0.0.2: bytes=32 time=1ms TTL=254
Reply from 10.0.0.2: bytes=32 time=1ms TTL=254
Reply from 10.0.0.2: bytes=32 time=1ms TTL=254

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

The taskbar at the bottom shows the Windows search bar, task view button, and several application icons. The system tray on the right displays the date and time as 09:38 on 17-08-2026, along with network and volume icons.



The screenshot shows a PC0 desktop environment. The 'Desktop' tab is selected in the top menu. A 'Command Prompt' window is open, displaying the following text:

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 10.0.0.2: bytes=32 time=1ms TTL=254
Reply from 10.0.0.2: bytes=32 time=1ms TTL=254
Reply from 10.0.0.2: bytes=32 time=1ms TTL=254

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=8ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 3ms

C:\>
```

The taskbar at the bottom shows the Windows search bar, task view button, and several application icons. The system tray on the right displays the date and time as 09:40 on 17-08-2026, along with network and volume icons.

SHETH L.U.J AND SIR MV COLLEGE

PRACTICAL NO:- 06

```
Router>
Router>enable
Router>configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 5
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router(config)#
Router(config)#crypto ipsec transform-set MYSET esp-aes esp-sha-hmac
Router(config)#
Router(config)#crypto map VPNMAP 10 ipsec-isakmp
*NOTICE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.1
Router(config-crypto-map)#set transform-set MYSET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#crypto map VPNMAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#exit
Router(config)#
Router(config)#end
Router#
*Aug 17, 09:45:25.4545: SYS-6-CONFIG_I: Configured from console by console
Router#
```

```
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 10.0.0.2: bytes=32 time=1ms TTL=254
Reply from 10.0.0.2: bytes=32 time=1ms TTL=254
Reply from 10.0.0.2: bytes=32 time=1ms TTL=254

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=8ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 3ms

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time=10ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126

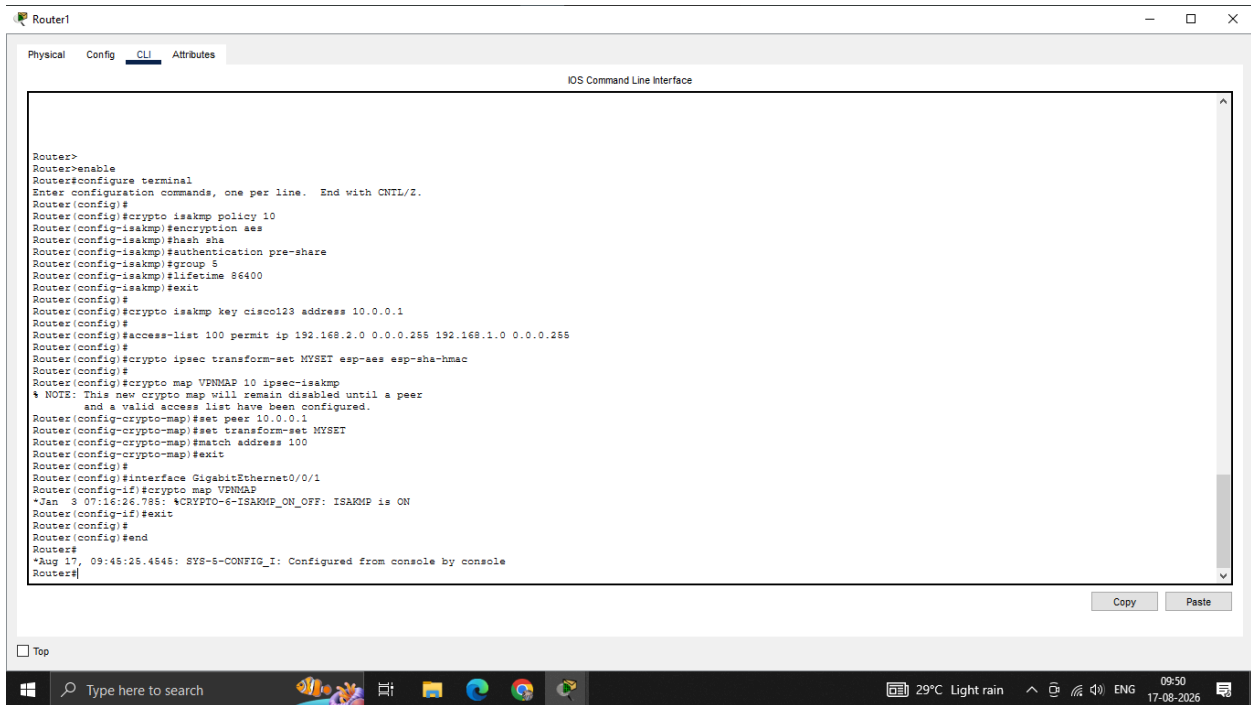
Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>
```

SHETH L.U.J AND SIR MV COLLEGE

PRACTICAL NO:- 06

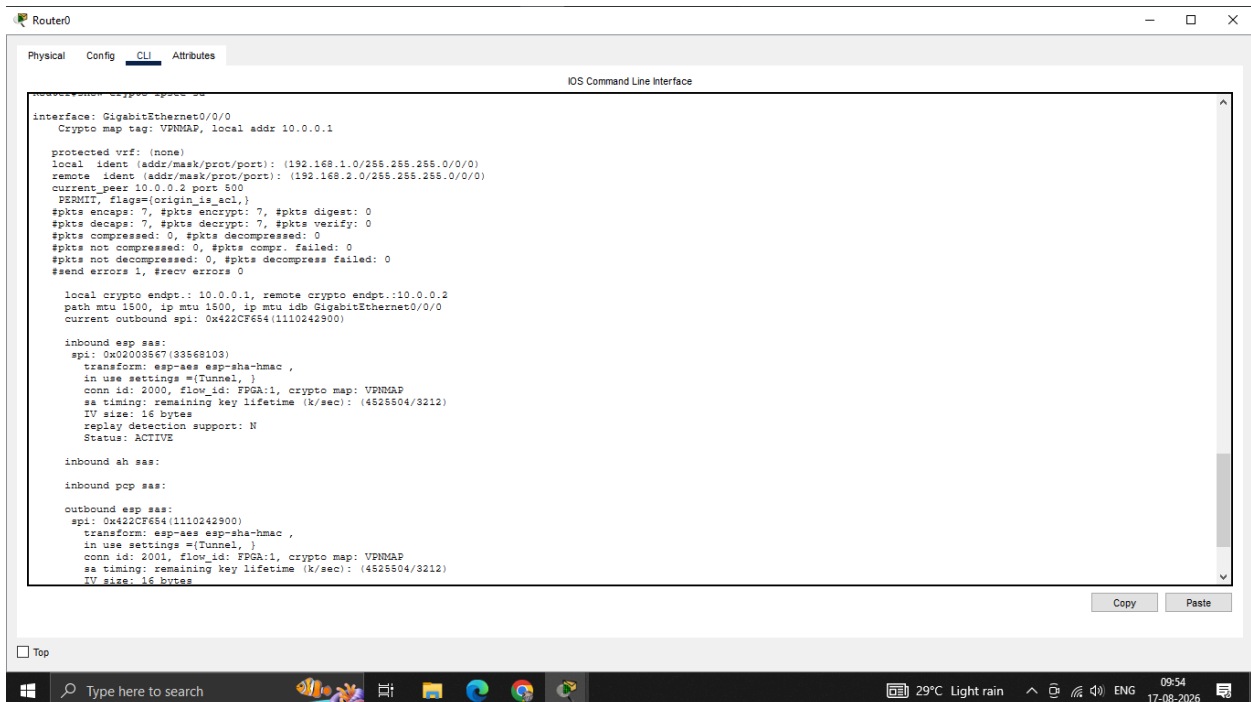
Router1 IPsec configuration



The screenshot shows the CLI of Router1 with the following configuration commands entered:

```
Router>
Router>enable
Router>configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encrptation aes
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 5
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#
Router(config)#crypto isakmp key cisco123 address 10.0.0.1
Router(config)#
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router(config)#
Router(config)#crypto ipsec transform-set MYSET esp-aes esp-sha-hmac
Router(config)#
Router(config)#crypto map VPNMAP 10 ipsec-isakmp
* NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.1
Router(config-crypto-map)#set transform-set MYSET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#crypto map VPNMAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router(config-if)#exit
Router(config)#end
Router#
*Aug 17, 09:45:25.4645: SYS-5-CONFIG_I: Configured from console by console
Router#
```

IPsec is successfully ACTIVE.



The screenshot shows the CLI of Router0 displaying the status of the IPsec configuration:

```
interface: GigabitEthernet0/0/0
Crypto map tag: VPNMAP, local addr 10.0.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
PERMIT, flags=(origin_is_acl, )
  #pkts encaps: 7, #pkts encrypt: 7, #pkts digest: 0
  #pkts decaps: 7, #pkts decrypt: 7, #pkts verify: 0
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 1, #recv errors 0

local crypto endpt.: 10.0.0.1, remote crypto endpt.: 10.0.0.2
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0/0
current outbound spi: 0x422CF654(1110242900)

inbound esp sas:
  spi: 0x02005567(33569103)
    transform: esp-aes esp-sha-hmac ,
    in use settings = (Tunnel, )
    conn id: 2000, flow_id: FPGA:1, crypto map: VPNMAP
    sa timing: remaining key lifetime (k/sec): (4525504/3212)
    IV size: 16 bytes
    replay detection support: N
    Status: ACTIVE

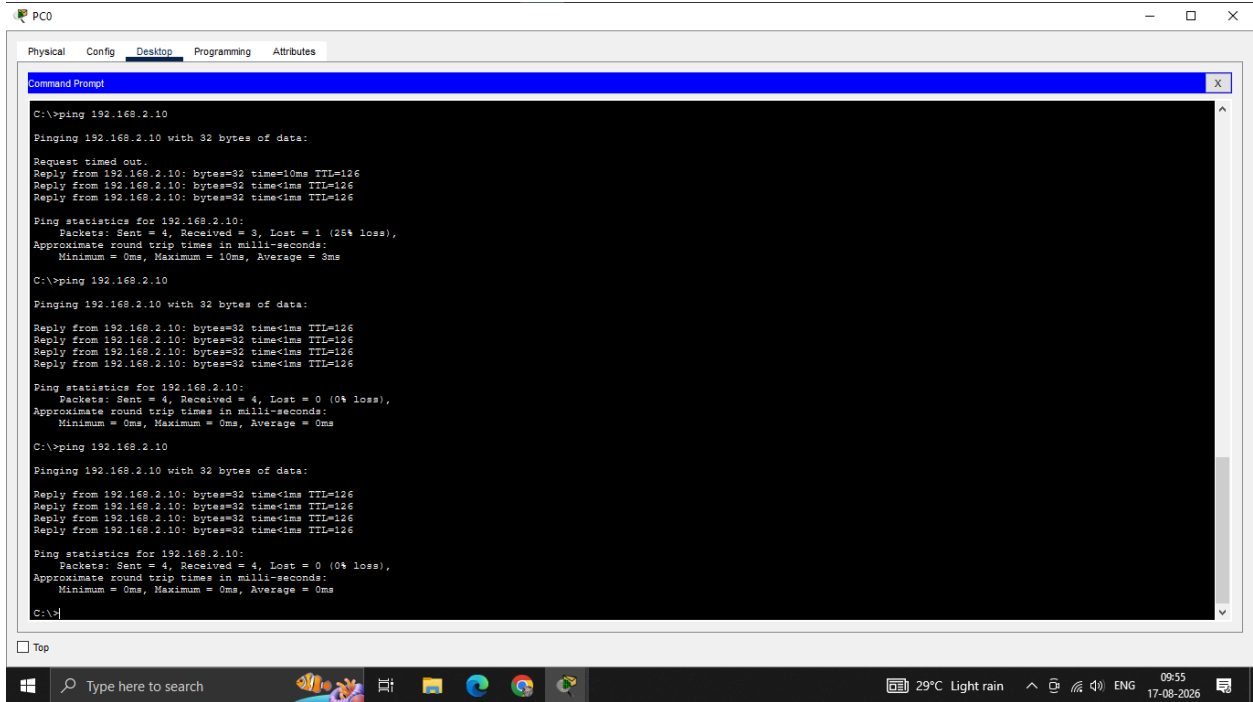
inbound ah sas:

inbound pcp sas:

outbound esp sas:
  spi: 0x422CF654(1110242900)
    transform: esp-aes esp-sha-hmac ,
    in use settings = (Tunnel, )
    conn id: 2001, flow_id: FPGA:1, crypto map: VPNMAP
    sa timing: remaining key lifetime (k/sec): (4525504/3212)
    IV size: 16 bytes
```

SHETH L.U.J AND SIR MV COLLEGE

PRACTICAL NO:- 06



The screenshot shows a Windows 10 desktop environment. A 'PC0' window is open, displaying a 'Command Prompt' window. The Command Prompt shows the results of a ping command to 192.168.2.10. The first ping attempt shows a 'Request timed out' for the first packet, followed by three successful replies. The second ping attempt shows four successful replies. The third ping attempt also shows four successful replies. The taskbar at the bottom includes the Start button, a search bar, and several application icons. The system tray on the right shows the date and time as 09:55 on 17-08-2026, along with weather information (29°C, Light rain) and language settings (ENG).

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time=10ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```